

Skeletal Transformer Encoder

Input Skeleton Sequence

Shape: $[B, T = 32, D = 117]$

Linear Projection + Sinusoidal PE

$\text{Linear}(117 \rightarrow 128) + \text{PE}(T = 32, d = 128) \rightarrow [B, 32, 128]$

$4 \times$ Pre-LN Layers

Multi-Head Self-Attention (MHSA)

Pre-LN $\rightarrow 8$ Heads, $d_k = 16$ ($d = 128$)

$\text{Softmax}(QK^T / \sqrt{d_k})V + \text{Residual Add}$

Feed-Forward Network (FFN)

Pre-LN $\rightarrow \text{Linear}(128 \rightarrow 256) \rightarrow \text{GELU}$

$\rightarrow \text{Linear}(256 \rightarrow 128) + \text{Residual Add}$

Temporal Global Average Pooling (GAP)

$\text{pooled} = \frac{1}{T} \sum_{t=1}^T z_t \rightarrow [B, 128]$

Two-Stage MLP Classifier Head

$\text{Linear}(128 \rightarrow 64) \rightarrow \text{ReLU} \rightarrow \text{Linear}(64 \rightarrow 22)$

Output: $P \in \Delta^{21}$ across 22 Exercise Classes